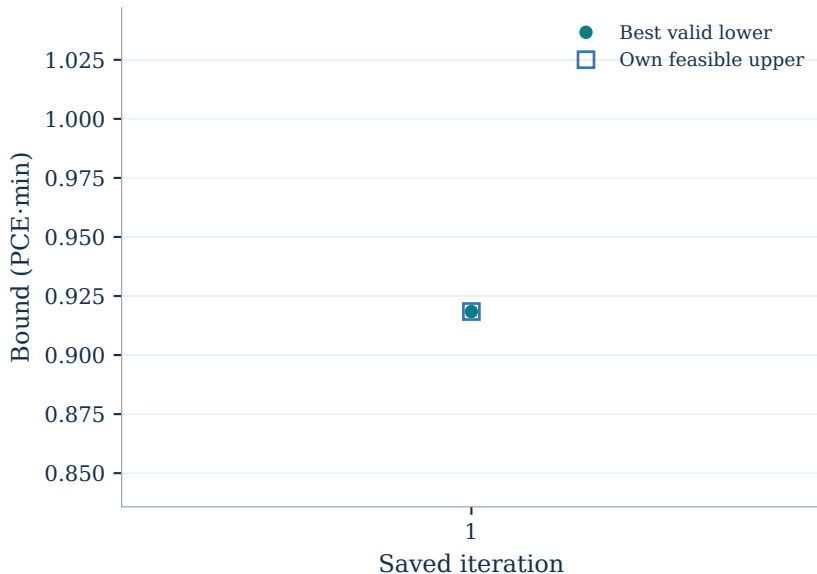


Lagrangian bounds and certified gap

a Valid bounds on the same graph



b Bound certificate

